Email & Network Indicator Investigation Helper

Technical Report | Version 1.0.0 | 10 November 2026

Project type: Defensive cybersecurity analysis tool
Data policy: Synthetic/offline input only

1. Abstract

This project implements a lightweight offline helper for investigating supplied email headers and DNS indicators. It extracts sender-related domains, records SPF/DKIM/DMARC results as stated in message headers, displays Received routing fields and parses supplied DNS records. The output is an evidence summary with cautious observations and explicit limitations. It does not claim that a message is malicious or benign and does not independently verify DNS or authentication.

2. Problem statement and objectives

Email investigations involve clues spread across message headers, DNS records and network observations. Analysts can over-trust visible sender fields, treat an authentication failure as proof of phishing, or infer an origin IP from an untrusted Received line. This project structures supplied evidence while preserving uncertainty.

• Parse a supplied .eml message using a standard email parser.

• Extract sender domains from From, Reply-To, Return-Path and Message-ID.

• Summarise supplied SPF/DKIM/DMARC results.

• Preserve Received header order and extract syntactically valid IP addresses.

• Parse supplied A, AAAA, MX, TXT, NS, PTR and CNAME records.

• Flag selected inconsistencies and missing evidence without asserting attribution.

• Provide repeatable tests and machine-readable JSON output.

3. Scope and assumptions

The tool is an offline parser. It assumes the analyst has permission to inspect the input and that supplied DNS text is already collected. It does not perform DNS resolution, SPF evaluation, DKIM signature verification, DMARC policy evaluation, IP reputation checks, packet dissection, malware analysis or mailbox access. The sample uses the reserved .test namespace and documentation IP ranges.

4. Design and implementation

The implementation uses Python 3.10+ standard-library modules: email.parser.BytesParser and email.policy for message structure; email.utils.parseaddr for mailbox extraction; regular expressions for token recognition; ipaddress for IP validation; argparse and json for CLI and output.

Stage	Description
Input	Analyst supplies .eml and optional DNS text file.
Parsing	Read selected fields and repeated headers.
Normalisation	Extract domains and validate IP address syntax.
Rule checks	Identify domain mismatches, authentication failures and missing evidence.
Output	Display structured JSON and optionally save report.json.

Header fields are read with a standard email parser. Authentication tokens are copied as reported; they are not independently authenticated. Received fields are preserved in supplied order rather than treated as a verified route. DNS input follows a simple name TYPE value format; unrecognised lines are retained as indicators with warnings.

5. Testing method and results

Eight unit tests cover a normal case, Reply-To mismatch, authentication failure, missing headers, invalid IP, malformed DNS line, absent DNS evidence and multiple Received fields. Run: `python -m unittest discover -s tests -v`. The recorded local run passed all 8 tests. Tests validate parser behaviour on fixtures, not real-world attribution or detection accuracy.

ID	Test case	Expected outcome
T01	Matching sender + auth pass	Three auth results; no mismatch
T02	Reply-To domain differs	Observation raised
T03	DMARC fail	Reported as observation
T04	Missing Received/auth	Missing-evidence notes
T05	Invalid IPv4	Rejected
T06	Malformed DNS line	Retained with warning
T07	No DNS input	No lookup claimed
T08	Multiple Received fields	Input order preserved

6. Security interpretation

A different Reply-To domain may be legitimate for support desks, mailing platforms or outsourced workflows. SPF/DKIM/DMARC results depend on the evaluated domain and whether Authentication-Results was inserted by a trusted receiving system. A Received chain can contain forged entries before a trusted boundary. An IP in a header is not automatically the sender's true origin. DNS observations are time-sensitive and should be timestamped and sourced.

MITRE ATT&CK's Phishing technique (T1566) can provide behavioural context when independent evidence supports phishing behaviour. Mapping an isolated header anomaly to ATT&CK is not proof of an attack.

7. Limitations

• No live DNS or reputation lookup.

• No cryptographic DKIM verification or SPF/DMARC recomputation.

• No PCAP parsing in the Python implementation.

• No determination of trusted Received boundaries.

• Simple DNS text format; not a full zone-file parser.

• Heuristic observations can produce false positives and false negatives.

• No guarantee of complete RFC coverage.

8. Ethical and safe-use considerations

Use synthetic samples or data the analyst is authorised to inspect. Wireshark sample captures must be selected with attention to their licence, terms and sensitivity. Never include credentials, personal data, confidential headers or unapproved targets in public examples or demo recordings. Scapy is not required for this version; if used in a future lab, restrict packet generation to an isolated local environment.

9. Future work

Potential extensions include graphical .eml upload, a validated PCAP workflow using Wireshark/tshark, timestamped DNS collection, configurable trusted relay boundaries, richer parsing tests, and HTML/PDF report export. Each extension should preserve source provenance and distinguish observed facts from analyst hypotheses.

10. References

• Wireshark, Wireshark Documentation, https://www.wireshark.org/docs/ (accessed 2026-11-10).

• Wireshark Wiki, Sample Captures, https://wiki.wireshark.org/SampleCaptures (accessed 2026-11-10).

• Scapy, Documentation, https://scapy.readthedocs.io/en/latest/ (accessed 2026-11-10).

• MITRE, MITRE ATT&CK, https://attack.mitre.org/ (accessed 2026-11-10).

• IETF, RFC 5322: Internet Message Format, https://www.rfc-editor.org/rfc/rfc5322 (accessed 2026-11-10).

• IETF, RFC 7208: Sender Policy Framework, https://www.rfc-editor.org/rfc/rfc7208 (accessed 2026-11-10).

• IETF, RFC 7489: DMARC, https://www.rfc-editor.org/rfc/rfc7489 (accessed 2026-11-10).

11. Conclusion

The project provides a small, reproducible way to structure supplied email and DNS evidence. Its value is transparent parsing and careful interpretation, not automated threat verdicts. Further packet-level integration should be added only with a documented sample capture and repeatable offline procedure.